Weichi Zhao

AI Infra / MaaS / LLM Serving Platform Engineer

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Target roles: LLM serving, OpenAI-compatible MaaS, AI gateway, Kubernetes model deployment, offline inference and data distillation.

Profile

AI Infra engineer focused on productionizing LLM serving end to end, from Kubernetes one-click model deployment to OpenAI-compatible MaaS, LiteLLM/Higress gateways, SGLang Router, offline inference, and data distillation. I work on public model services, serving-framework adaptation, model gateways, routing reliability, agent-client integration, and heterogeneous GPU/XPU environments, with a focus on turning runnable models into deliverable, governable, observable, and recoverable platform capabilities.

Impact Highlights

- Supported a production data-distillation path at around 10M requests per day with success rate above 99%, covering task sharding, concurrency control, failure recovery, and result writeback.
- Validated an offline batch inference -> SGLang Router -> multi-backend serving path for Qwen3: about 300K requests / 9.75B tokens, about 99.9% success rate, and below 5% platform/Router CPU and memory overhead.
- Worked on SGLang/vLLM/Dynamo serving tuning across KV/prefix cache, PD disaggregation, cache-aware/bucket routing, and QPS/TTFT/Tok/s validation.
- Built OpenAI-compatible MaaS public-service capabilities on top of LiteLLM/Higress-style open-source gateway components, covering multi-model access, authentication, rate limiting, model permissions, usage tracking, and troubleshooting.
- Applied agent-assisted workflows in current MaaS/LLM serving work for code search, incident writeups, design drafts, release checks, and knowledge-base updates, turning repeated manual steps into reusable iteration loops.
- Expanded Kubernetes-native one-click model deployment lifecycle coverage across CRDs/controllers, Service/Endpoint readiness, Router discovery, metrics Services, finalizers, multi-cluster rollout, and rollback paths.

Selected IP & Honors

- Co-inventor on granted invention patents for elastic model inference serving and cache-aware deep-learning training.
- Contributor to authorized software copyrights for cloud/HPC workflow platforms; team-related honors include China Youth May Fourth Medal collective honor, National Worker Pioneer recognition, and Gordon Bell Prize team context.

Core Skills

LLM Serving	SGLang, vLLM, OpenAI-compatible APIs, streaming, embeddings, Responses API, PD disaggregation, KV/prefix cache, cache-aware and bucket routing.
Model Deployment	Kubernetes-native inference workloads, operators/controllers, CRDs, Helm, Kustomize, Service/Endpoint readiness, finalizers, rollout safety, retry and rollback workflows.
MaaS Public Service	Higress, LiteLLM, AI Gateway, API keys, team/model access, TPM/RPM limits, token budgets, model mapping, observability.
Agent Platform	Claude Code / Codex / OpenCode integration, tool-calling API compatibility, OpenAI/Anthropic-style adapters, long-context cache optimization, agent-assisted engineering iteration.
Engineering	Go, Python, Linux, Prometheus metrics, MySQL/SQLite/Redis, distributed systems debugging, performance and capacity testing.

Experience

Zhejiang Lab, High-Performance Computing Facility Research Center — 2024 - Present

Senior Research Specialist, AI Infra and MaaS platform

- Built and operated LLM serving platform capabilities covering public model-service entrypoints, OpenAI-compatible APIs, Higress/LiteLLM gateway integration, SGLang/vLLM framework adaptation, Kubernetes-native one-click model deployment, SGLang Router, offline batch inference, and Agent-facing API compatibility.
- Owned production troubleshooting and hardening for model deployment lifecycle, Router worker registration convergence, Service/Endpoint readiness, PD prefill/decode topology, CrashLoop and long-connection failures, gateway header/model mapping issues, and multi-cluster rollout validation.
- Designed MaaS control-plane and optimizer-loop mechanisms around LiteLLM, Router, offline batch inference, and gateway metrics, with public-model limit adjustment, offline concurrency tuning, observation windows, rollback criteria, and online-SLA-first rules.

Zhejiang Lab, Intelligent Supercomputing Research Center — Aug 2021 - 2023

Engineering Specialist, cloud-native AI and HPC platforms

- Worked on cloud-native AI/HPC platform projects, including Kubernetes adaptation on domestic OS and accelerators, device plugin customization for accelerator availability, PyTorch/model adaptation, and distributed training framework exploration.
- Supported model training and inference platform adaptation in domestic heterogeneous hardware/software environments, covering job submission, resource discovery, container runtime validation, and baseline performance verification.

Selected AI Infra Projects

OpenAI-compatible offline batch inference platform — **LLM batch inference**

Asynchronous JSONL batch execution for large-scale, non-real-time model workloads.

- Designed Batch API compatible capabilities: file upload, batch lifecycle, cancellation, result/error files, streaming file processing, request-level error isolation, and backend concurrency control.
- Refactored the platform toward multi-replica operation with Kubernetes leader election, sharded workers, header forwarding, upload limits, stale-file cleanup, Kustomize upgrades, monitoring, and failure-tolerant finalization.

- Supported production data-distillation and batch-generation workloads, later reaching daily processing around 10M requests with success rate above 99% under controlled concurrency.
- Validated an offline batch inference -> SGLang Router -> multi-backend path for Qwen3-235B-A22B-Thinking-2507 at about 300K requests, about 9.75B tokens, about 99.9% success rate, and below 5% platform/Router CPU/memory overhead.

Kubernetes-native one-click model deployment and serving reliability

Model deployment, tuning, routing, and orchestration

Standardized model images, launch parameters, resources, ports, health checks, Services, worker registration, and API validation so models move from configuration to request reachability with less manual YAML work.

- Worked on Kubernetes-native inference-task lifecycle orchestration across model images, launch parameters, resource specs, replica topology, Service/Endpoint readiness, Router discovery, metrics Services, finalizers, and rollback paths, reducing manual deployment and troubleshooting cost.
- Participated in SGLang/vLLM/Dynamo research and adaptation validation, mapping PagedAttention/RadixAttention, KV cache, prefix cache, PD disaggregation, multi-level cache reuse, context length, concurrency, batching, and cache-hit-rate tuning boundaries.
- Improved Router worker synchronization by reconciling desired workers against live `/workers` state and requeueing on missing, extra, or failed registrations.
- Enhanced inference-instance startup and readiness with Pod-IP early registration, `publishNotReadyAddresses`, preserved Service finalizers, Router metrics Service generation, and instance-level auto-restart with retry budgets.
- Validated cache-aware/bucket scheduling strategies and diagnosed ARG_MAX overflow, Router not managed after Worker Ready, stale Service selectors, and long-connection timeout behavior.

MaaS public model service and agent integration optimization

OpenAI-compatible API / Agent runtime

Unified model access, authentication, rate limiting, protocol compatibility, cost governance, and troubleshooting for internal users, agent clients, and offline workloads on top of LiteLLM/Higress-style gateway capabilities.

- Built OpenAI-compatible public model-service entrypoints covering Chat Completions, Responses-style APIs, embeddings, Batch API, and agent-client access while hiding differences across model backends, routers, and heterogeneous clusters from business callers.
- Contributed public-service governance around API keys, teams, model access, quotas, TPM/RPM limits, token budgets, model mapping, and cost-control policies for shared model access.
- Analyzed protocol compatibility boundaries for Claude Code / Codex-style clients, including system prompts, tool schemas, tool results, streaming events, Responses-style APIs, and OpenAI/Anthropic field differences.
- Diagnosed low SGLang prefix/KV cache hit rates for Claude Code long-context requests by comparing rendered prompts, cached-token logs, and token LCP; identified dynamic attribution fields that polluted request prefixes and mitigated the issue.
- Embedded agent tools into cross-repository navigation, log summarization, design drafts, PR/issue writing, release checks, knowledge-base updates, and MaaS onboarding troubleshooting.

Kubernetes device plugin and heterogeneous accelerator platform

GPU/XPU scheduling

Cloud-native resource discovery, scheduling, and validation for domestic heterogeneous accelerator environments.

- Customized device-plugin and resource-management paths for accelerator availability, node resource reporting, container runtime validation, card-level scheduling, and failure isolation.
- Built deployment, upgrade, rollback, and E2E validation procedures for Kubernetes AI/HPC environments on domestic operating systems and accelerator stacks.

Earlier Projects

Arizona State University, SNAC Lab — Jan 2020 - Aug 2020
Virtual platform developer and research assistant

- Developed THOTH Lab modules for SDN/OpenFlow labs, Boto3-backed AWS control, and an Android cybersecurity research prototype.

Arizona State University, CIDSE — Aug 2019 - Dec 2020
Graduate service assistant

- Assisted CSE205, CSE412, CSE434, and CSE464.

Education

Arizona State University — May 2019 - Dec 2020
Master of Computer Science, GPA 3.97/4.00

Arizona State University — Aug 2015 - May 2019
Bachelor of Science in Computer Science, GPA 3.40/4.00

Last updated: July 2026